

Soru 2)

x	0,2	0,4	0,6	0,8	1,0
f(x)	0,975362	0,917212	0,8010330	0,6386093	0,3943235

a) $f'(0,4)$ ve $f''(0,4)$ türevlerini yukarıdaki hesaplamaya göre mümkün durumlar

$f'(0,4)$ için 1. adım $\Rightarrow$ 3 nokta ya da nokta

$$h=0,2$$

$$f'(x_0) = \frac{1}{2h} [-3f(x_0) + 4f(x_0+h) - f(x_0+2h)]$$

$$f'(0,4) = \frac{1}{0,4} [-3f(0,4) + 4f(0,2) - f(0)] = -0,393425$$

2. adım $\Rightarrow$ 3 nokta ya da nokta $h=0,2$

$$f'(x_0) = \frac{1}{2h} [-3f(0,4) + 4f(0,2) - f(0)] = \text{hesaplamaya gerekmez}$$

$f(0)$ değerini bilmiyoruz

3. adım $\Rightarrow$ 3 nokta orta nokta

$$f'(x_0) = \frac{1}{2h} [f(x_0+h) - f(x_0-h)] \quad h=0,2$$

$$f'(0,4) = \frac{1}{0,4} [f(0,6) - f(0,2)] = 0,425562$$

• $f''(0,4)$ için $\Rightarrow$ 5 nokta orta nokta

$$f''(x_0) = \frac{1}{h^2} [f(x_0-h) - 2f(x_0) + f(x_0+h)] \quad h=0,2$$

$$f''(0,4) = \frac{1}{0,04} [f(0,2) - 2f(0,4) + f(0,6)] = -1,13097$$

①

sur 3)

$n=4$ için

$$x_4 = x_3 - \left(\frac{(x_3 - x_2) \cdot (2x_3 + 3\cos x_3 - e^{x_3})}{2x_3 + 3\cos x_3 - e^{x_3} - 2x_2 - 3\cos x_2 + e^{x_2}} \right)$$

$$x_4 = 1,216410 - \left(\frac{0,005285}{-0,217225} \right)$$

$$x_4 = 1,24068$$

$$|x_4 - x_3| = 0,02427 > 10^{-5} \text{ (DEVAM)}$$

$n=5$ için

$$x_5 = x_4 - \left(\frac{(x_4 - x_3) \cdot (2x_4 + 3\cos x_4 - e^{x_4})}{2x_4 + 3\cos x_4 - e^{x_4} - 2x_3 - 3\cos x_3 + e^{x_3}} \right)$$

$$x_5 = 1,24068 - \left(\frac{-0,000100}{-0,102957} \right)$$

$$x_5 = 1,23920$$

$$|x_5 - x_4| = 0,00057 > 10^{-5}$$

$n=6$ için

$$x_6 = x_5 - \left(\frac{(x_5 - x_4) \cdot (2x_5 + 3\cos x_5 - e^{x_5})}{2x_5 + 3\cos x_5 - e^{x_5} - 2x_4 - 3\cos x_4 + e^{x_4}} \right)$$

$$x_6 = 1,23920 - \left(\frac{-6,181726 \cdot 10^{-2}}{0,042072} \right)$$

$$x_6 = 1,23969$$

$$|x_6 - x_5| = 0,0000013 < 10^{-5}$$

$$K \leq k \Rightarrow x_6 = 1,23969$$

sonuç ②

Soru 1)

$$\ln(x-1) + \cos(x-1) = 0$$

$$f'(x) = \frac{1}{x-1} - \sin(x-1) = 0 \quad x_0 = 1,3$$

$$\frac{n=1}{x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}} = 1,3 - \frac{f(1,3)}{f'(1,3)} = 1,3 - \frac{(0,268636)}{3,032812} = 1,381847$$

$$\frac{n=2}{x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}} = 1,381847 - \frac{f(1,381847)}{f'(1,381847)} = 1,381847 - \frac{(-0,026286)}{2,246213} = 1,397320$$

$$|1,397320 - 1,381847| = 0,01547 > 10^{-5}$$

Devam

n=3

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 1,397320 - \frac{f(1,397320)}{f'(1,397320)} = 1,397320 - \frac{(-0,000911)}{2,125913} = 1,3977481$$

$$|1,3977481 - 1,397320| = 0,00042 > 10^{-5}$$

Devam

n=4

$$x_4 = x_3 - \frac{f(x_3)}{f'(x_3)} = 1,3977481 - \frac{(-0,000001)}{2,126510} = 1,3977485$$

$$|1,3977485 - 1,3977481| = 4 \times 10^{-7} < 10^{-5}$$

$$Kök = 1,3977485$$

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örü2)

b) $f'(0,6)$ ve $f''(0,6)$ yaklaşık hesaplayın tüm mümkün durumlar
değerini bul

* $f'(0,6)$ için $\rightarrow$ 1. adım 3 nokta ve nokta $h=0,2$

$$f'(0,6) = \frac{1}{0,4} [-3f(0,6) + 4f(0,8) - f(1,0)] = \underline{0,635125}$$

$\rightarrow$ 2. adım 3 nokta orta nokta $h=0,2$

$$f'(0,6) = \frac{1}{0,4} [f(0,8) - f(0,4)] = \underline{-0,697904}$$

$\rightarrow$ 3. adım 3 nokta orta nokta $h=0,2$

$$f'(0,6) = \frac{1}{0,4} [f(0,8) - f(0,4)] = \underline{0,697904}$$

$\rightarrow$ 4. adım 5 nokta orta nokta $h=0,2$

$$f'(x_0) = \frac{1}{12h} [f(x_0-2h) - 8f(x_0-h) + 8f(x_0+h) - f(x_0+2h)]$$

$$f'(0,6) = \frac{1}{2,4} [f(0,2) - 8f(0,4) + 8f(0,8) - f(1,0)] = \underline{-0,682305}$$

* $f''(0,6)$ için 1. adım $h=0,2$ için orta nokta

$$f''(0,6) = \frac{1}{(0,2)^2} [f(0,4) - 2f(0,6) + f(0,8)] = \underline{-1,432392}$$

$\rightarrow h=0,4$ için orta nokta

$$f''(0,6) = \frac{1}{(0,4)^2} [f(0,2) - 2f(0,6) + f(1,0)] = \underline{-1,543983}$$

sovr 6)

$$f(x) = e^{3x} - 4x$$

$$f'(x) = 3e^{3x} - 4$$

$$f''(x) = 9e^{3x}$$

$$f'''(x) = 27e^{3x}$$

$$h_0 = \sqrt[3]{\frac{3\varepsilon}{M_3}}$$

$$\varepsilon = 0,5 \times 10^{-4}$$

$$M_3 = \max_{[0,3,0,7]} |f'''(x)|$$

$$M_3 = \max |27e^{3x}| = |27e^{3 \cdot 0,7}|$$

$$M_3 \approx 220,486587$$

$$h_0 = \sqrt[3]{\frac{3 \cdot 0,5 \cdot 10^{-4}}{220,486587}}$$

$$h_0 \approx 0,008795$$

soru 4)

$$\int_0^2 x^2 \ln(x^2+1) dx \quad h=0.25$$

a) Birleştirilmiş yamuk

$$0.25 = \frac{2}{n}$$
$$n=8$$

$$I = \frac{1}{8} [f(0) + 2f(0.25) + 2f(0.5) + 2f(0.75) + 2f(1) + 2f(1.25) + 2f(1.5) + 2f(1.75) + f(2)]$$

$$I = \frac{1}{8} \cdot [25.27577]$$

$$I \approx 3.15942$$

$$f(0) = 0$$

$$2f(0.25) = 0.00752$$

$$2f(0.5) = 0.11157$$

$$2f(0.75) = 0.50207$$

$$2f(1) = 1.38625$$

$$2f(1.25) = 2.91057$$

$$2f(1.5) = 5.30394$$

$$2f(1.75) = 8.58101$$

$$f(2) = 6.43725$$

b) Birleştirilmiş Simpson

$$0.25 = \frac{2}{n} \quad I = \frac{1}{12} [f(0) + 4f(0.25) + 2f(0.5) + 4f(0.75) + 2f(1) + 4f(1.25) + 2f(1.5) + 4f(1.75) + f(2)]$$
$$n=8$$

$$I = \frac{1}{12} \cdot [32.31223]$$

$$I \approx 3.10935$$

$$f(0) = 0$$

$$4f(0.25) = 0.01512$$

$$2f(0.5) = 0.11156$$

$$4f(0.75) = 1.00914$$

$$2f(1) = 1.38628$$

$$4f(1.25) = 5.88114$$

$$2f(1.5) = 5.30394$$

$$4f(1.75) = 17.1723$$

$$f(2) = 6.43725$$

sayfa ①

Soru 4)

$$\int_0^2 x^2 \ln(x^2+1) dx \quad h=0.25$$

a) Birleştirilmiş yamuk

$$0.25 = \frac{2}{n}$$

$n=8$

$$I = \frac{1}{8} [f(0) + 2f(0.25) + 2f(0.5) + 2f(0.75) + 2f(1) + 2f(1.25) + 2f(1.5) + 2f(1.75) + f(2)]$$

$$I = \frac{1}{8} \cdot [25.27577]$$

$$I \approx 3.15947$$

$$f(0) = 0$$

$$2f(0.25) = 0.00757$$

$$2f(0.5) = 0.11157$$

$$2f(0.75) = 0.50207$$

$$2f(1) = 1.38629$$

$$2f(1.25) = 2.94057$$

$$2f(1.5) = 5.30384$$

$$2f(1.75) = 8.58601$$

$$f(2) = 6.43775$$

b) Birleştirilmiş Simpson

$$0.25 = \frac{2}{n}$$

$n=8$

$$I = \frac{1}{12} [f(0) + 4f(0.25) + 2f(0.5) + 4f(0.75) + 2f(1) + 4f(1.25) + 2f(1.5) + 4f(1.75) + f(2)]$$

$$I = \frac{1}{12} \cdot [32.31223]$$

$$I \approx 3.10935$$

$$f(0) = 0$$

$$2f(0.25) = 0.00757$$

$$2f(0.5) = 0.11157$$

$$2f(0.75) = 0.50207$$

$$2f(1) = 1.38629$$

$$2f(1.25) = 2.94057$$

$$2f(1.5) = 5.30384$$

$$2f(1.75) = 8.58601$$

$$f(2) = 6.43775$$

sayfa ①

$$c) \quad 0,25 = \frac{2}{n} \quad \int_0^2 2 \ln(x^2+1) dx$$

$$n=8$$

$$\int_0^b f(x) dx = 2h \sum_{j=0}^{n/2} f(x_{2j})$$

$$I = 0.5 [f(0,25) + f(0,75) + f(1,25) + f(1,75)]$$

$$I = 0.5 [6,018119]$$

$$I \approx 3,00906$$

$$f(0,25) = 0,003739$$

$$f(0,75) = 0,251036$$

$$f(1,25) = 1,420256$$

$$f(1,75) = 4,293008$$

sayfa 2

sovr 3) $f(x) = 2x + 3\cos x - e^x$
 $f(x_n) = 2x_n + 3\cos x_n - e^{x_n}$

$$\begin{matrix} x_0 = 1 \\ x_1 = 2 \end{matrix}$$

$$f(x_0) = 0,09026250$$

$$f(x_1) = -4,63749660$$

$$x_n = x_{n-1} - \frac{(x_{n-1} - x_{n-2}) \cdot (2x_{n-1} + 3\cos x_{n-1} - e^{x_{n-1}})}{(2x_{n-1} + 3\cos x_{n-1} - e^{x_{n-1}}) - 2x_{n-2} - 3\cos x_{n-2} + e^{x_{n-2}}}$$

n=2 için

$$x_2 = x_1 - \left(\frac{(x_1 - x_0) \cdot (2x_1 + 3\cos x_1 - e^{x_1})}{2x_1 + 3\cos x_1 - e^{x_1} - 2x_0 - 3\cos x_0 + e^{x_0}} \right)$$

$$x_2 = 2 - \left(\frac{1 \cdot (4 - 1,2448440 - 7,389056)}{4 - 1,2448440 - 7,389056 - 2 - 1,620906 + 2,718281} \right)$$

$$x_2 = 1,162925$$

$$|x_2 - x_1| < 10^{-5} \text{ değil}$$

$$1,162925 - 2 = 0,837075 > 10^{-5} \text{ (DEVAM)}$$

n=3 için

$$x_3 = x_2 - \left(\frac{(x_2 - x_1) \cdot (2x_2 + 3\cos x_2 - e^{x_2})}{2x_2 + 3\cos x_2 - e^{x_2} - 2x_1 - 3\cos x_1 + e^{x_1}} \right)$$

$$x_3 = 1,162925 - \left(\frac{-0,264368}{4,35408} \right)$$

$$x_3 = 1,216410$$

$$|x_3 - x_2| = 0,053485 > 10^{-5} \text{ (DEVAM)}$$

Sayfa ①